REVEAL: Agentic In-Hand Manipulation for Language-Guided Target Presentation

Anonymous CVPR submission



Figure 1. **REVEAL overview.** New visual evidence guides semantic decisions as the hand inspects and presents a requested object face. This staged simulation illustration is not a recorded rollout or physical-robot result.

Abstract

Language-guided in-hand manipulation requires a robot to find a requested object face, reorient it while preserving contact, and continue acting while visual reasoning is pending. We present REVEAL, an asynchronous system that connects a visual agent to a history-conditioned motor policy through a shared goal bank. The agent uses updated images to inspect and bind semantic targets, while teacher-supervised local controllers adapt joint commands during reasoning and execution. We evaluate complete systems across six object-specific simulation releases and two semantic mappings, with separate cube interventions for control and feedback. Under the same maximum call budget, updated images yield 32/40 completions versus 9/40 when the initial image is repeated. Changed-mapping comparisons further support system-level adaptation within the tested releases. Independent confirmation does not establish an advantage of learned waiting control over fixed slowing. Multi-object physical-hand evaluation remains pending.

1. Introduction

A command such as "present the face marked T" specifies a visual goal, not a sequence of hand motions. The robot must find the mark, associate it with a controllable orientation, and rotate the object while preserving contact. The target may be hidden by the object or the fingers, so an informative action also changes the grasp and the next observation. In-hand target presentation therefore couples visual reasoning with contact-rich control.

Vision-language agents offer a flexible way to interpret such requests, but their decisions arrive asynchronously. During inference, the hand and object continue to move. Repeating the previous command may lose the grasp or leave the target occluded; slowing motion alone may preserve contact without producing a useful correction. A practical system

needs a local controller that responds to current measurements while semantic reasoning is pending.

Our key insight is to separate the rate of semantic decisions from the rate of motor control while keeping both interfaces grounded in the same goal representation. An asynchronous visual agent inspects the object and issues a validated semantic command. A history-conditioned student maps the active goal and measured pose to joint targets. Two compact local controllers adapt those targets during pending requests and ordinary execution, using privileged teacher actions for training. This lets the hand respond to its current state without waiting for another high-level call.

We instantiate this design by adapting a ViserDex-style RGB-to-pose policy [1] to a continuous semantic goal input while reusing compatible recurrent weights. Our Astra route emits public tool arguments that map to the shared goal bank; it does not expose hidden model embeddings. A fine-tuned Qwen3-VL baseline predicts a goal vector directly, and a rule-based VLM parses a color and readability decision. These interfaces provide a common motor policy while retaining the differences between an agent with feedback and history and single-step grounding baselines.

We evaluate complete systems across six object-specific simulation releases and two semantic mappings. Separate interventions use only the marked cube to isolate feedback and control mechanisms. REVEAL completes 35/40 cube trials, compared with 30/40 for the fine-tuned VLM and 17/40 for the rule baseline. Continued feedback and fresh images improve completion in matched cube studies. A separate dice layout experiment tests adaptation to a changed visual mapping. The results support the value of updated visual evidence and system-level layout adaptation; they do not establish a learned waiting-control advantage or zero-shot transfer of one policy across objects. Multi-object physical-hand evaluation remains pending.

Our contributions are threefold. First, we formulate language-guided target presentation as asynchronous semantic grounding coupled to continuous in-hand control through a shared goal interface. Second, we introduce teacher-supervised local controllers that adapt joint commands using measured state while high-level inference is pending and during execution. Third, we provide six-object simulation comparisons and matched cube interventions that separate the effects of feedback from controller components, with explicit evidence limits for unresolved gains and pending hardware evaluation.

We freeze task mappings, checkpoints, stopping rules, and physical success criteria before evaluation. Each system retains 40 raw trials, with service exclusions applied symmetrically within comparisons. We report physical reach separately from correct semantic binding and autonomous completion.

2. Related Work

Visual dexterous manipulation. ViserDex uses monocular visual pose estimation and recurrent control for dexterous reorientation, with Gaussian representations supporting visual training [1]. We reuse this division between perception and control and adapt a released Revo3 policy to semantic goals. ManipTrans studies residual learning for dexterous bimanual transfer [4]. These works motivate a reusable motor policy and residual adaptation. Our residual parameterization alone is not a novelty claim. The question here is how a local policy should condition its behavior on the phase and intention of an asynchronous semantic agent.

Language and visual agents. Magma connects multimodal representations to agent actions [9]. VERA translates visual plans into robot actions and includes language-conditioned dexterous in-hand reorientation [5]. Its hand experiments include directional rotation prompts. Consequently, language-guided in-hand rotation is not an unoccupied task category. Our setting emphasizes identifying a requested marked face under occlusion, maintaining contact during external reasoning, and checking a canonical target at autonomous stopping. Those distinctions require direct evidence, rather than a claim of being the first language-controlled hand.

Asynchronous robot policies. Real-time chunk execution [2], online corrections for VLA chunks [7], and latency-aware RL [10] address inference delays through different execution and learning designs. DEFLECT studies delay-robust asynchronous VLAs through temporal counterfactual preferences [11]. LiMA combines sparse semantic intention and asynchronous action generation for dexterous manipulation [3]. These are close methodological references. Our current implementation learns small teacher-supervised controllers around a separate recurrent policy and public semantic tools. A controlled comparison with a generic time-aware residual policy is still needed to show that the active semantic goal and request anchor add value beyond reactive correction.

3. Task and Evaluation Contract

Semantic target. Q1 covers six object-specific simulation releases: a marked cube, a wooden die, a duck, a hedgehog, a pig, and a milk carton. Q2/Q3 interventions use only the marked cube. Cube instructions identify one letter from $\{T, E, R, L, S, H\}$; the die instructions request one through six dots. Each target has a canonical world orientation $Q(g)$ that places its marked face upward. The environment scores the full canonical quaternion, including its fixed twist, rather than accepting every orientation with the correct normal. The language prompt is therefore less specific than the scoring

convention. This convention is disclosed and held constant for all methods. A face-normal-only score can be reported as an additional diagnostic without replacing the primary metric.

At step t, the agent observes RGB evidence I_t and the instruction ℓ. The motor system observes proprioception x_t and measured pose $\hat{p}_t$. True pose p_t, teacher actions, and future semantic responses are unavailable to online control. The simulator exposes them only for training labels and independent scoring. We denote the active goal vector by g_t, recurrent state by h_t, and the pending-request flag by m_t.

Physical time. One request may be pending at a time. The default protocol allows six model calls per episode and 3,600 logical control steps at 30 Hz. If a call starts at k and takes L wall-clock seconds, its response becomes usable only after the call completes and at least $\lceil 30L \rceil$ control steps have elapsed. Physics continues throughout this interval. The simulator currently runs slower than real time. This protocol represents inference delay in simulated physical time, but does not demonstrate a 30 Hz hardware pipeline.

Autonomous completion. All cube systems share a measured-pose stopping gate, evaluated at 3 Hz. It requires a declared binding for the requested symbol, no pending request, a non-inspection state, and measured full-orientation error at most $\epsilon = 0.16$ radians. The gate may stop independently of an agent's native stop suggestion. The dice extension uses a calibrated completion correction for Astra and a measured-pose gate for the bare baselines, with the same independently scored tolerance; its protocol is documented separately in the supplement. At stop time τ, the independent success variable is

$$S = \mathbf{1}\left[\hat{b}_\tau = b^*(\ell),\ d(q_\tau, Q(b^*)) \le \epsilon,\ \neg D_{0:\tau}\right], \quad (1)$$

where $\hat{b}_\tau$ is the declared face binding, q_τ is true orientation, $D_{0:\tau}$ indicates an earlier terminal drop, and $d(q, Q) = 2\arccos(|\langle q, Q\rangle|)$ for normalized quaternions. Merely passing through a target does not count. No extra hold duration is added after stopping. A pose false stop and a semantically wrong binding are failures even if the hand retains the object. The frozen drop rule is distance of at least 0.24 metres from the nominal object grasp position.

Matched reporting. Every arm uses 40 raw trials from two seeds, with 20 trials per seed and matching initial physical states and scored physical targets. Instructions also match within the control and feedback studies. In the remapping study, the requested letter changes with the mapping to preserve the physical target. A service error or measured model latency above 120 seconds is recorded as a service exclusion. For a panel of methods $\mathcal{C}$, the eligible set removes the union of excluded trial identifiers across $\mathcal{C}$. We report both raw and eligible outcomes, never replace an excluded trial, and never mix denominators from different panels. Long service delays remain visible without being interpreted as controller failures.

4. Method

The system has a semantic interface, a visual motor policy, and local adaptation. The semantic interface changes the active intention. The local controllers change how that intention is executed using current measurements.

4.1. Reusing a visual pose-conditioned student

A frozen CNN predicts the image coordinates and depth of the object center and eight keypoints from a 120×120 RGB crop. Calibration and rigid fitting recover the measured position and XYZW orientation, together with a fit residual. Cube trials use rendered object views with live hand occlusion; deployment receives no ground-truth pose.

The inherited student expects geometric goals. We replace its goal channels with a learned projection of $g_t \in \mathbb{R}^{768}$:

$$z_t = \tanh(W_g g_t + b_g) \in \mathbb{R}^{64}, \quad (2)$$

$$(u_t, h_{t+1}) = \pi_\theta([x_t, z_t], \hat{p}_t, h_t). \quad (3)$$

Here $x_t \in \mathbb{R}^{109}$ contains target-free proprioceptive features and history, while $u_t \in \mathbb{R}^{21}$ gives normalized joint-target proposals. Compatible encoders, recurrent weights, actor weights, and normalization statistics are initialized from the released policy. Neither target quaternion nor relative-goal quaternion is an input to the migrated student.

We fit the new goal interface and adapt the recurrent core with frozen-teacher actions in a DAgger-style procedure [6]. Teacher state and actions are training-only. The cube deployment uses the frozen RGB pose estimator; object-specific policy and perception differences for the other releases are documented in the supplement.

4.2. Semantic grounding and public actions

The cube uses six normalized goal vectors $B = \{g_1, \ldots, g_6\}$ from cached SigLIP2 text features [8]; each is bound to a color skill and canonical asset orientation.

Semantic agent interface. Given the requested symbol and current image, the hosted agent emits a validated public action (inspect, reorient, verify, stop, or abstain). A color argument selects a vector from B; inspection may change the motor goal without binding the requested symbol. Each request spans 10–90 control steps, after which updated visual evidence and action history inform another decision. Public arguments map deterministically to the student goal; hidden agent features are not used.

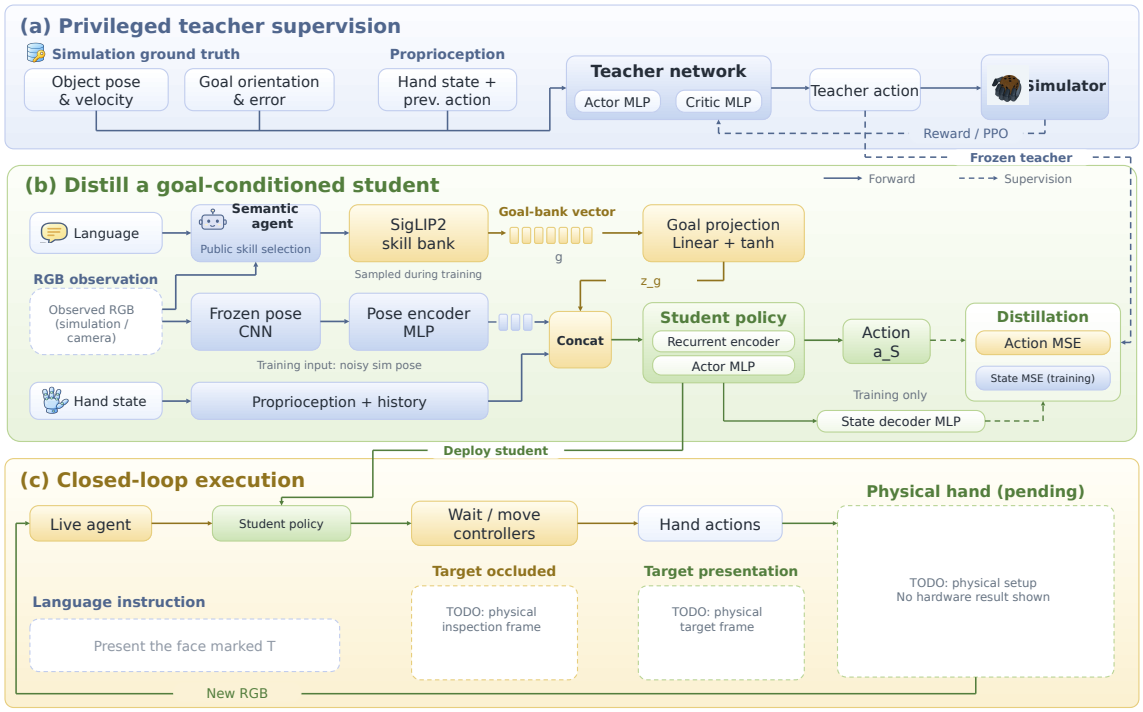


Figure 2. **REVEAL training and deployment pipeline.** Privileged supervision trains a student conditioned on a public goal-bank vector. The semantic agent selects skills, and local controllers adapt their execution. Dashed physical-hand slots are pending; they do not report hardware results.

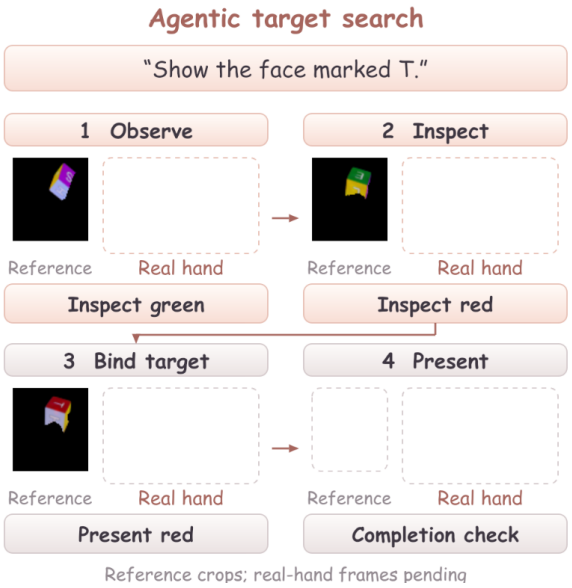


Figure 3. **Visual target search.** The agent observes, inspects candidate faces, binds the requested target, and issues a presentation command. The reference crops are simulated; real-hand frames are pending.

4.3. Learning control while a request is pending

An outstanding query does not determine whether the current movement is still safe. We construct a deployable feature vector s_t from proprioception, measured pose, student belief, proposed action, pose-fit residual, semantic binding status, and request status. It also includes the measured pose and command at the request anchor and elapsed pending time. The active goal g_t is encoded separately. Before a binding is established it denotes the startup or inspection skill, not the unavailable future target.

A sensor encoder and a goal encoder feed a fusion layer. The waiting actor predicts a speed logit r_t and a residual $v_t \in \mathbb{R}^{21}$. Let $\bar{q}_t$ be normalized current joint position and $\bar{u}_t$ the clipped student proposal. The applied command is

$$\alpha_t = \mathrm{sigmoid}(4r_t + \mathrm{logit}(0.15)), \tag{4}$$

$$b_t = \begin{cases} \bar{q}_t + \alpha_t(\bar{u}_t - \bar{q}_t), & m_t = 1, \\ \bar{u}_t, & m_t = 0, \end{cases} \tag{5}$$

$$a_t = \mathrm{clip}\big(b_t + 0.35\tanh(v_t), -1, 1\big). \tag{6}$$

Zero outputs reproduce fixed slowing with coefficient 0.15 while a query is pending. The environment's joint mapping and smoothing remain unchanged. The residual permits contact adjustments outside the interpolation between current joints and the student proposal.

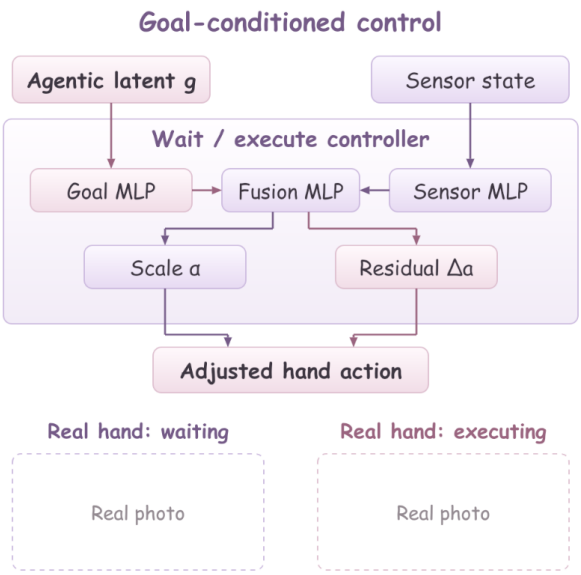


Figure 4. **Local control component.** A goal-conditioned actor combines sensor and goal features to scale the student proposal and add a bounded residual. The photo slots in the source illustration are pending and do not show real-hand results.

Waiting targets are generated by the frozen teacher with the request anchor as the holding objective. This supervision is available in simulation. The deployed actor receives measurements and the active goal only. It never receives the answer being computed. The objective fits the applied action to the teacher label and penalizes residual magnitude:

$$\mathcal{L}_{\text{control}} = \mathbb{E}_{(s,g,a^*)\sim\mathcal{D}}\left[\|a_\phi(s,g) - a^*\|_2^2 + 0.01\|\delta_\phi(s,g)\|_2^2\right]. \tag{7}$$

Refit protocol, checkpoint selection, and independent confirmation are reported in the supplement. Only pending-request outputs are deployed; the student, predictor, execution controller, and stopping gate remain fixed.

4.4. Learning execution after a decision

A second actor uses the student proposal as its base and learns a bounded teacher-supervised correction for non-pending execution. Its active goal is explicit, although the goal-input interventions do not establish a benefit from this pathway (Sec. 5).

4.5. Closed-loop execution

At each control step, the system updates visual pose, advances the recurrent student, chooses the waiting or execution actor, and applies the resulting joint targets. A completed semantic response is validated before it changes the active skill or declared binding. The next image reflects the motion that occurred while the previous response was pending. The common stopping gate is checked separately from semantic action generation. Invalid model output cannot supply ground-truth information or silently select the requested target.

Student adaptation and controller learning are evaluated separately. A new student changes the belief and proposal distribution seen by the controllers. Their checkpoint-bound weights therefore require compatibility validation before transfer.

5. Experiments

We evaluate REVEAL through four questions. Q1 tests complete systems across objects and semantic mappings; Q2 studies local control during deliberation; Q3 tests visual feedback; Q4 concerns physical-hand evaluation. Q2 and Q3 use the marked cube.

Q1. How do complete systems compare across objects and semantic mappings?
Q2. How should the hand act during deliberation?
Q3. Does continued visual feedback help?
Q4. Does the system work on physical objects?

5.1. Setup and reporting

Experiments use Revo3 cube and dice assets in Isaac Lab 3.0/Newton with validated contact capacities. Each agent condition has two seeds and 40 scheduled trials. Initial physical states are matched within each study. Within the cube studies, all systems share the frozen pose estimator, goal bank, latent student, 0.16-radian success threshold, and measured-pose stopping gate. Dice trials share an object-specific student and noisy simulator pose; Astra additionally uses calibrated completion correction. We report these backend differences rather than pooling their protocols. The primary fine-tuned and rule VLM baselines omit the added waiting and execution actors. Astra uses adaptive inspection, whereas the local baselines use a fixed unreadable-view schedule. This is a comparison of complete systems, with natural model latencies, rather than an isolated test of model weights.

The latest cube system replaces only its waiting actor. Release cohorts and module interventions remain separate. Table percentages use common service-eligible trials within the stated comparison. All raw records are retained, and no failed or excluded slot is replaced. Wilson intervals describe individual rates. The new waiting contrasts and six registered visual contrasts use Holm correction within their respective families. The independent confirmation and image-refresh tests form a separate two-test Holm family. Historical paired tests remain exploratory and unadjusted.

5.2. Q1: Systems across objects and mappings

We compare original and changed semantic mappings across six object-specific releases. Earlier native-image cohorts are retained separately in the supplement. Those cube and die

cohorts do not establish Astra superiority over fine-tuning (paired $p = 0.267$ and $p = 0.4545$), and are not pooled with the mapping study.

Four additional objects. The registered duck, hedgehog, pig and milk-carton releases complete all 960 raw trials (four objects, three methods, two mappings, forty trials). Table 1 reports the mapping cohorts for all six objects. Full Astra has the highest observed rate in each new-object cell. Its changed-mapping rates are 60.0%, 80.0%, 82.5% and 27.5%, compared with 5.0%, 5.0%, 7.5% and 2.5% for fine-tuning. These are object-specific system comparisons, not a common-policy transfer test. Three historical service failures are excluded symmetrically from their original-mapping comparisons. Milk-carton control remains difficult: Astra drops in 15/40 original and 20/40 changed-mapping trials. Higher completion therefore does not establish uniformly stable grasping. The supplement records the numerical reset review and sensitivity analysis.

Visual-semantic intervention. We keep predictor input and lighting fixed while changing the upper-level image. Identity, one cyclic letter–color permutation, and letter masking are evaluated on 40 matched states per method. All three conditions use geometry-consistent reconstructed face images, including identity. Consequently, this is a controlled image benchmark rather than the untouched RGB main comparison.

Astra retains 32/40 successes after permutation, while fine-tuned success falls from 30/40 to 16/40 (Holm-adjusted $p = 0.0130$ across six planned contrasts). Rule success remains 23/40. Equal aggregate rates do not establish equivalence or identical trajectories. Masking removes almost all completion, supporting dependence on visible semantic evidence. Nine of eleven fine-tuned wrong-binding trials after permutation select the original letter–color association. This is consistent with reliance on learned associations, but does not identify an exclusive memorization mechanism. One cube and one permutation do not establish broad semantic generalization.

Dice layout transfer. A separate 240-trial study rotates only the native visual mesh, preserving collision and inertia. The same two layouts use a prompt distinguishing printed counts from motor-skill IDs. With 40 matched trials per cell, Astra completes 28/40 identity and 26/40 permuted trials, fine-tuning 23/40 and 0/40, and rules 13/40 and 1/40. There are no service exclusions. In the permuted condition, Astra's exploratory paired contrasts give $p = 2.98 \times 10^{-8}$ against fine-tuning and $p = 5.96 \times 10^{-8}$ against rules. Its own layout contrast ($p = 0.688$) does not establish equivalence. The local baselines receive current RGB and instruction, whereas Astra also uses probes, history and noisy feedback. This result supports system-level layout transfer, not isolated model superiority. The supplement reports protocol differences, binding errors and a recorded probing example.

5.3. Q2: Controller ablations on the cube

On the marked cube, we compare four ways of acting while a semantic request is pending: continue the student proposal, interpolate toward it with fixed speed 0.15, learn that speed, and learn both speed and a bounded joint correction. Every condition uses the same frozen execution actor outside pending intervals. Astra, the student, visual pose predictor, stopping gate, and physical horizon are fixed.

The speed-only actor is trained with zero waiting residual, matching full-actor data, initialization, optimizer, sampling and 2,400 updates. Reproduction yields identical deployed full-actor tensors. Held-out checks verify identical non-pending actions.

Continued execution succeeds in 25/40 trials, fixed slowing in 30/40, learned slowing in 28/40, and learned slowing with joint correction in 34/40 (Table 3). There are no service exclusions. The full actor has the highest observed success, but learning only the speed does not improve on fixed slowing. Holm-adjusted exact paired p-values for the three registered adjacent contrasts are 0.604, 0.804, and 0.539. This cohort does not resolve those component gains.

Total drops are twelve, four, eight, and five. Full control has four later-pending drops versus one for learned slowing. Fixed slowing has four startup drops and no later drops. Highest completion therefore does not coincide with fewest drops. The supplement separates phases. Shared execution weights do not imply identical states or subsequent agent decisions.

Trials use seeds 185501 and 185502 with reversed condition order for the second seed. Checkpoints are fixed before these outcomes. Historical training-version comparisons remain separate from this structural ablation.

Independent confirmation. Without refitting, a new two-seed cohort gives 25/39 successes for continuation, 31/39 for fixed slowing, and 30/39 for learned slowing with correction. One service slot is excluded symmetrically, with all 120 raw trials retained. Learning versus fixed slowing has six versus seven exclusive successes (Holm $p = 1.0$). The learned arm has no waiting drops but eight execution drops, compared with three and two for fixed slowing. This confirmation does not establish a learned-control advantage. Later failures may reflect states and commands induced by waiting, despite identical execution weights. The supplement reports the full panel.

Other component evidence. Earlier waiting–execution factorial tests do not establish a reliable interaction. Explicit

Object	Perception	Original mapping			Changed mapping		
		Rule	Fine-tuned	Full Astra	Rule	Fine-tuned	Full Astra
Cube	RGB pose	23/40 (57.5)	30/40 (75.0)	32/40 (80.0)	23/40 (57.5)	16/40 (40.0)	32/40 (80.0)
Die	Noisy pose	13/40 (32.5)	23/40 (57.5)	28/40 (70.0)	1/40 (2.5)	0/40 (0.0)	26/40 (65.0)
Duck	Noisy pose	15/38 (39.5)	14/38 (36.8)	25/38 (65.8)	9/40 (22.5)	2/40 (5.0)	24/40 (60.0)
Hedgehog	Noisy pose	17/39 (43.6)	13/39 (33.3)	28/39 (71.8)	10/40 (25.0)	2/40 (5.0)	32/40 (80.0)
Pig	Noisy pose	21/40 (52.5)	16/40 (40.0)	37/40 (92.5)	15/40 (37.5)	3/40 (7.5)	33/40 (82.5)
Milk carton	Noisy pose	13/40 (32.5)	9/40 (22.5)	17/40 (42.5)	3/40 (7.5)	1/40 (2.5)	11/40 (27.5)

Table 1. **Q1: system comparison across six objects and two semantic mappings.** Entries give successes / common service-eligible trials (percent), retaining 40 raw trials per cell. Fine-tuned and rule baselines omit the added control modules. Each object uses its own frozen student and adaptation. The cube uses reconstructed upper-level images and RGB pose, the die rotates its visual mesh, and the four new objects permute surface letters in native RGB with noisy simulator pose. Within each object, both mappings share physical assets and weights. These mapping cohorts are separate from the native-image release comparisons. No cross-release pooled average or zero-shot object-transfer claim is made.

System	Identity	Permuted	Masked
Astra full	32/40	32/40	0/39
Fine-tuned VLM, bare	30/40	16/40	3/40
Rule VLM, bare	23/40	23/40	0/40

Table 2. **Q1: cube semantic-mapping comparison.** Identity and permutation retain all 40 trials. Astra masking has one service exclusion. Its paired masking contrast uses 31/39 identity successes versus 0/39 masked successes. The local methods have no exclusions. This study uses the original Astra actors.

Waiting policy	Success	%	Drops	Wait drops
Continue student	25/40	62.5	12	7
Fixed slowdown	30/40	75.0	4	4
Learned slowdown	28/40	70.0	8	1
Learned slowdown + correction	34/40	85.0	5	4

Table 3. **Q2: cube controller ablation.** Each condition has 40 scheduled trials. All arms use the same frozen controller outside pending requests. Successes and drops use the common service-eligible panel. Wait drops combine startup and later pending intervals. All 40 trials per arm are service-eligible.

controller-goal interventions also do not establish a benefit, and a separate learned execution-residual gate fails its development admission rule. These negative results and the matched-state diagnostic are retained in the supplement. The new waiting study changes neither the execution actor nor its goal input.

5.4. Q3: Visual feedback on the cube

On the same cube, we compare Astra under two observation protocols. One receives the initial RGB observation and measured feedback, then commits to a complete plan containing one to six bounded actions. It receives no later semantic observation. Its last non-abstain skill can continue until the shared stopping gate or horizon terminates the trial.

Protocol	Success	%	Drops	Calls
One initial plan	6/39	15.4	0	40
Full feedback	33/39	84.6	5	120

Table 4. **Q3: initial plan versus feedback on the cube.** Each arm has 40 scheduled trials. The full-feedback condition is shared with Q2, rather than counted as another 40 raw trials. Calls include all raw episodes. One service-affected slot is excluded symmetrically. Different inference budgets are part of the protocol comparison.

The other can use up to six calls with updated observations and history. Both retain continuous pose feedback to the student, the same full waiting and execution actors, and the same stopping rule.

The initial plan succeeds in 6/39 eligible trials, compared with 33/39 for continued feedback. Twenty-seven paired trials succeed only with feedback and none only with the initial plan (exact $p = 1.49 \times 10^{-8}$). Raw success is 6/40 versus 34/40. One initial-plan HTTP 503 timeout is excluded from both arms without replacement. Actual calls total 40 and 120.

The initial plan abstains in 28 eligible trials, of which 22 succeed with feedback. Among eligible feedback trials, 36 contain inspection, 32 first bind after inspection, and 31 successes use multiple calls. These traces support obtaining evidence before binding, without demonstrating correction of an executed wrong binding. The protocols differ in inference budget. Zero initial-plan drops accompany frequent abstention, not better completion. The supplement retains a recorded example and the earlier cohort.

Does the next image matter? A separate cohort retains the same six-call cap and trigger policy in both arms, changing only the upper agent's image. One arm receives current RGB, the other the exact first frame on every call. Numeric pose feedback, action history, lower visual perception and control remain live. Forty pairs share initial physical states and first images, with no service exclusions.

Upper RGB	Success	%	Drops	Calls
Current image	32/40	80.0	3	112
Repeated first frame	9/40	22.5	4	98

Table 5. **Q3: fresh visual evidence on the cube.** Forty trials per arm, with the same call cap and frozen local controllers. Only upper-level RGB is frozen. Calls are realized totals, not matched compute.

Physical object	Full Astra	Fine-tuned VLM	Rule VLM
Cube	[TODO: data]	[TODO: data]	[TODO: data]
Die	[TODO: data]	[TODO: data]	[TODO: data]
Duck	[TODO: data]	[TODO: data]	[TODO: data]
Hedgehog	[TODO: data]	[TODO: data]	[TODO: data]
Pig	[TODO: data]	[TODO: data]	[TODO: data]
Milk carton	[TODO: data]	[TODO: data]	[TODO: data]

Table 6. **Q4: planned physical-hand evaluation.** Cells marked TODO await matched physical trials and audit; the simulation results in Table 1 do not fill these entries.

Current images yield 32/40 completions versus 9/40 for repeated first frames (Table 5). All 23 discordant pairs favor current images, a 57.5-point gain (exact $p = 2.38 \times 10^{-7}$, Holm $p = 4.77 \times 10^{-7}$). Abstentions fall from 24 to two, while drops change from four to three. Transport hashes verify the delivered images. This supports the value of later visual evidence with other feedback retained. Actual calls, tokens and trajectories differ, so equal maximum calls do not establish equal realized compute.

5.5. Q4: Multi-object hardware evaluation

Hardware protocol. **[TODO: Run and audit the multi-object physical-hand evaluation.]** The planned protocol compares the complete system and two baselines on each available object. Report completion, drops, false stops, inference cost, calibration and synchronized videos. No physical-hand outcomes are available yet. The supplementary figures show the planned presentation protocol and qualitative panel layout; their real-hand image regions remain unfilled.

Supporting cube perception study. On 6,144 rendered observations, RGB-D reaches 64.73% pose recall versus 37.53% for RGB at 10 mm/10°. This independent estimator comparison uses ideal target masks and virtual depth; it does not establish downstream task gains or physical-camera accuracy. Full protocol and occlusion analysis are in the supplement.

6. Discussion and Limitations

Q2/Q3 examine mechanisms only on the marked cube, so their effects are not established across objects. The image-refresh intervention retains numeric feedback, history and lower perception. Its call cap is matched, but realized compute and trajectories differ.

The earlier waiting study favors joint correction by point estimate, but independent confirmation gives 30/39 versus 31/39 for fixed slowing. Zero waiting drops in that learned arm coexist with eight execution drops. A completion advantage remains unconfirmed. Explicit controller goals provide no demonstrated benefit. The rejected execution gate is not deployed. Native-layout Astra–fine-tuned differences remain unresolved on both objects.

Q1 tests one changed mapping per object, with system differences in history, feedback and control. Cube perception uses rendered RGB, while the other five objects use noisy simulator pose and separate weights. These results establish neither real-camera robustness nor zero-shot object transfer. Hosted-model revisions limit replication. Predictor accuracy does not establish downstream gains; Q4 remains pending.

7. Conclusion

We studied semantic in-hand reorientation with continuous control around asynchronous visual decisions. Cube interventions support continued feedback, and system comparisons support layout transfer on six tested objects. Component gains remain unresolved. The milk carton remains difficult, with only 27.5% completion after remapping despite the highest rate among the compared systems.

References

[1] Arjun Bhardwaj, Maximum Wilder-Smith, Mayank Mittal, Vaishakh Patil, and Marco Hutter. ViserDex: Visual sim-to-real for robust dexterous in-hand reorientation. *arXiv preprint arXiv:2604.11138*, 2026. 2

[2] Kevin Black, Manuel Y. Galliker, and Sergey Levine. Real-time execution of action chunking flow policies. *arXiv preprint arXiv:2506.07339*, 2025. 2

[3] Ning Chen, Yankai Fu, Junkai Zhao, Qianpu Sun, Guocai Yao, Pengwei Wang, Zhongyuan Wang, and Shanghang Zhang. LiMA: Bridging long-term imagination to real-time dexterous manipulation via asynchronous diffusion. *arXiv preprint arXiv:2609.28431*, 2026. 2

[4] Kailin Li, Puhao Li, Tengyu Liu, Yuyang Li, and Siyuan Huang. ManipTrans: Efficient dexterous bimanual manipulation transfer via residual learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6991–7003, 2025. 2

[5] Sizhe Lester Li, Evan Kim, Xingjian Bai, Tong Zhao, Tao Pang, Max Simchowitz, and Vincent Sitzmann. Turning video models into generalist robot policies. *arXiv preprint arXiv:2605.27817*, 2026. 2

[6] Stephane Ross, Geoffrey Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, pages 627–635, 2011. 3

[7] Kohei Sendai, Maxime Alvarez, Tatsuya Matsushima, Yutaka Matsuo, and Yusuke Iwasawa. Leave no observation behind: Real-time correction for VLA action chunks. *arXiv preprint arXiv:2509.23224*, 2025. 2

[8] Michael Tschannen, Alexey Gritsenko, Xiao Wang, Muhammad Ferjad Naeem, Ibrahim Alabdulmohsin, Nikhil Parthasarathy, Talfan Evans, Lucas Beyer, Ye Xia, Basil Mustafa, Olivier Hénaff, Jeremiah Harmsen, Andreas Steiner, and Xiaohua Zhai. SigLIP 2: Multilingual vision-language encoders with improved semantic understanding, localization, and dense features. *arXiv preprint arXiv:2502.14786*, 2025. 3

[9] Jianwei Yang, Reuben Tan, Qianhui Wu, Ruijie Zheng, Baolin Peng, Yongyuan Liang, Yu Gu, Mu Cai, Seonghyeon Ye, Joel Jang, Yuquan Deng, and Jianfeng Gao. Magma: A foundation model for multimodal AI agents. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14203–14214, 2025. 2

[10] Brian Zhu, Momen Khalil, E Harrison, Emanuele Poggi, Philipp Schmitt, Bernd Kast, Philine Meister, Pranav Atreya, Qiyang Li, Finn Ferchau, Cesar Colmenero, Yash Shahapurkar, Gokul Narayanan, Melih Erdogan, Kai Wurm, Georg von Wichert, Oier Mees, Eugen Solowjow, Andrew Wagenmaker, and Sergey Levine. Learning to act while waiting: RL finetuning of generalist robot policies under inference latency. *arXiv preprint arXiv:2608.23831*, 2026. 2

[11] Yixiang Zhu, Yonghao Chen, Jiaxiang Zou, Yusong Hu, and Xinyu Chen. DEFLECT: Temporal counterfactual preference learning for delay-robust asynchronous VLAs. *arXiv preprint arXiv:2605.19294*, 2026. 2

LLM Usage Statement

Generative AI tools assisted with research discussion, implementation, experiment orchestration, literature retrieval, data analysis, and manuscript drafting. Astra and Qwen3-VL are also components evaluated in the research. Numerical results in this draft are derived from recorded experiments. Unfinished experiments are explicitly marked TODO. The authors must verify the final text, references, code, and results and remain responsible for the submitted work.